Open Problems on Strong Orientations of Digraphs

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Based in part on joint work with **Jørgen Bang-Jensen**



Strong orientations

k-strong connectivity

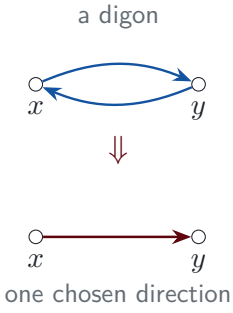
$|V(D)| \geq k+1$, and $D - S$ is strongly connected whenever $|S| < k$.

An orientation of a digraph

Retain every single arc.
Choose one direction from each digon.

Our objective

Preserve **k-strong connectivity** when orienting the digons.



The Jackson–Thomassen conjecture

Conjecture 1 (Jackson–Thomassen, 1989)

Every $2k$-strong digraph admits a k-strong orientation.

$$2k\text{-strong} \quad \Longrightarrow \quad k\text{-strong orientation?}$$

- The case $k = 1$ is known.
- Even for $k = 2$, no fixed connectivity threshold is known for general digraphs.

What can additional degree assumptions guarantee?

Two sharp degree conditions

Theorems (Bang-Jensen–W., 2026+)

Fix $k \geq 1$ and let n be sufficiently large. Either condition below guarantees a k-strong orientation of D.

Minimum semi-degree

$$\delta^0(D) \geq \left\lfloor \frac{n+k-1}{2} \right\rfloor$$

$\delta^0(D) = \min\{\delta^+(D), \delta^-(D)\}$.

Woodall-type condition

$$d_D^+(x) + d_D^-(y) \geq n + 2k - 2$$

For all distinct x, y with $xy \notin A(D)$.

Both bounds are best possible.

From Woodall-type to Meyniel-type Conditions

Woodall-type Theorem

$$d_D^+(x) + d_D^-(y) \geq n + 2k - 2$$

For every missing arc xy.

Meyniel-type Conjecture

$$d_D(x) + d_D(y) \geq 2n + 4k - 5$$

For every nonadjacent pair x, y.

Here $d_D(v) = d_D^+(v) + d_D^-(v)$, and nonadjacent means $xy, yx \notin A(D)$.

A Meyniel-type conjecture

Conjecture 2 (Bang-Jensen–W., 2026+)

Let $k \geq 1$ and $n \geq 2k+1$. Suppose D is an n-vertex $(2k-1)$-**strong** digraph and

$$d_D(x) + d_D(y) \geq 2n + 4k - 5$$

for every pair of nonadjacent vertices x, y. Then D admits a k-**strong orientation**.

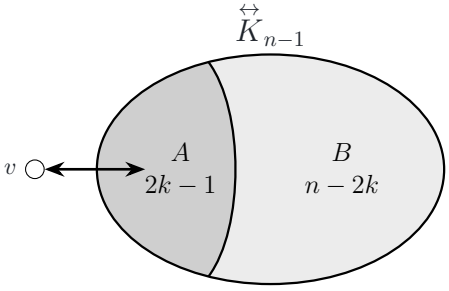
For $k = 1$, the degree bound becomes $2n - 1$: Meyniel's Hamiltonicity condition.

The Meyniel-type condition is best possible

Take a complete symmetric digraph on $n-1$ vertices. Add v, joined by digons to exactly $2k-1$ vertices.

For every nonneighbor $y \in B$ of v,

$$\begin{aligned} d_D(v) + d_D(y) &= 2(2k-1) + 2(n-2) \\ &= 2n + 4k - 6. \end{aligned}$$



In any orientation H,

$$d_H^+(v) + d_H^-(v) = 2k-1 \quad \Longrightarrow \quad \min\{d_H^+(v), d_H^-(v)\} \le k-1.$$

D is $(2k-1)$-strong, but no orientation is k-strong.

The semicomplete case

Semicomplete digraph: Every pair of vertices is adjacent.

Conjecture 3 (Bang-Jensen–Jordán, 2010)

Every $(2k-1)$-strong semicomplete digraph on at least $2k+1$ vertices contains a spanning k-**strong tournament**.

Meyniel-type conjecture $\Longrightarrow$ semicomplete conjecture.

- Known for $k = 1, 2$, and for $k = 3$ when $n \geq 9$.
- The connectivity bound $2k-1$ is best possible.

Open problems

Conjecture 1 (Jackson–Thomassen, 1989)

Every $2k$-strong digraph admits a k-strong orientation.

Conjecture 2 (Bang-Jensen–W., 2026+)

Every $(2k-1)$-strong digraph D on $n \geq 2k+1$ vertices satisfying $d_D(x) + d_D(y) \geq 2n + 4k - 5$ for every nonadjacent pair x, y admits a k-strong orientation.

Conjecture 3 (Bang-Jensen–Jordán, 2010)

Every $(2k-1)$-strong semicomplete digraph on at least $2k+1$ vertices contains a spanning k-strong tournament.

Conjecture 2 implies Conjecture 3.

Thank You!

Merci beaucoup !